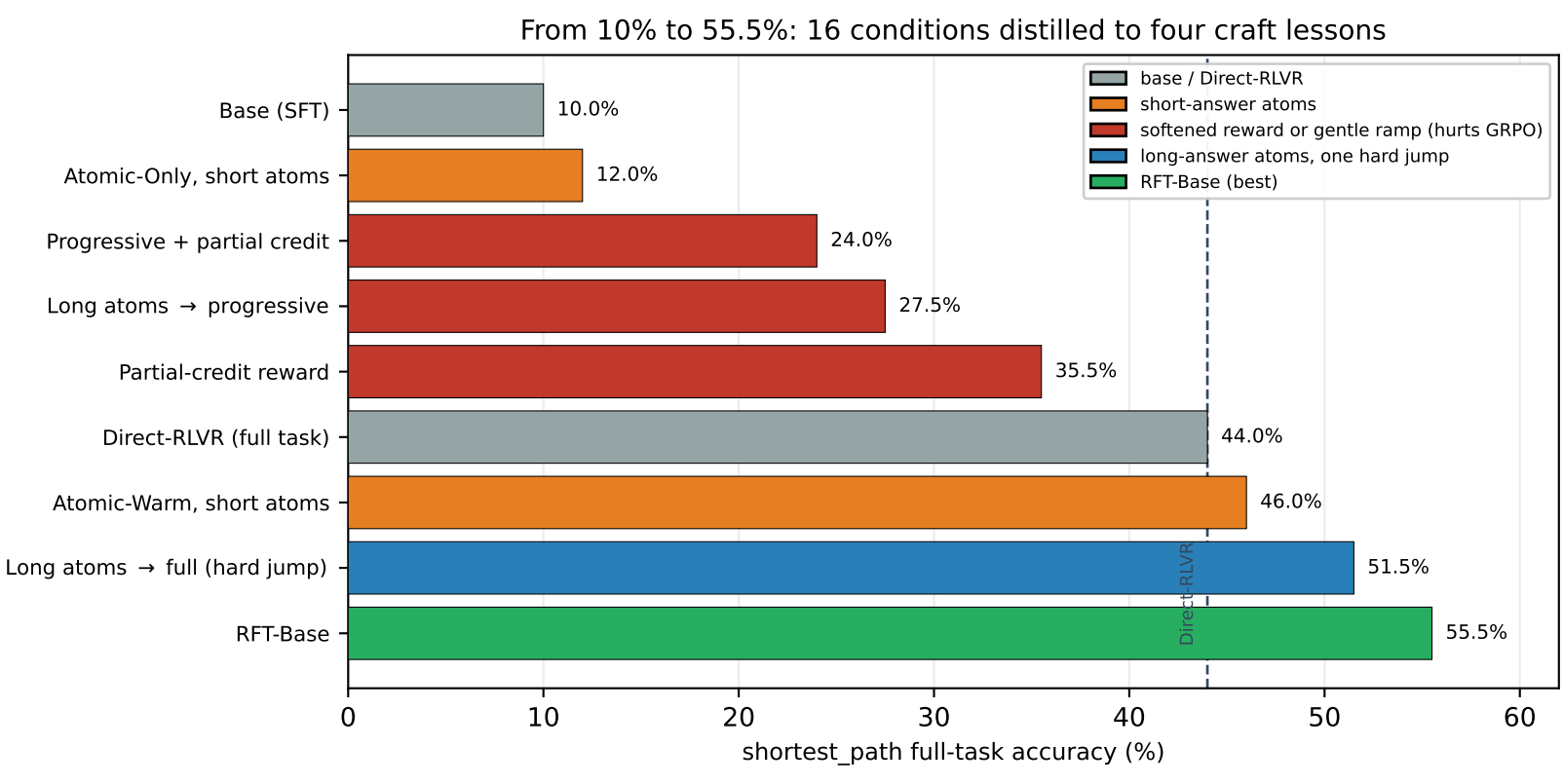
From 10% to 55.5%: 16 conditions distilled to four craft lessons
base / Direct-RLVR
short-answer atoms
softened reward or gentle ramp (hurts GRPO)
long-answer atoms, one hard jump
RFT-Base (best)
Base (SFT)
10.0%
Atomic-Only, short atoms
12.0%
Progressive + partial credit
24.0%
Long atoms → progressive
27.5%
Partial-credit reward
35.5%
Direct-RLVR (full task)
44.0%
Atomic-Warm, short atoms
46.0%
Long atoms → full (hard jump)
51.5%
RFT-Base
55.5%
Direct-RLVR
0
10
20
30
40
50
60
shortest_path full-task accuracy (%)